

# Explainable Hybrid Ensemble Model for Road Accident Severity Prediction Using Stacking

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## Abstract

Accident severity prediction plays a crucial role in improving road safety by tackling the global issue of reducing fatalities and economic losses in transportation systems. Challenges such as imbalanced datasets and the demand for interpretable models in safety-critical domains present significant obstacles. This paper introduces a hybrid ensemble model that integrates LightGBM, Support Vector Machine (SVM), and Random Forest, enhanced by a novel feature augmentation method inspired by hybrid quantum-classical techniques. The model was assessed on a balanced subset of around 31,245 accident records, drawn from a larger real-world dataset of 284,807 records, achieving a prediction accuracy of 93.36% for severity levels (Minor, Severe, Fatal), outperforming existing methods. To ensure transparency, Explainable AI (XAI) methods, specifically Local Interpretable Model-agnostic Explanations (LIME), were applied, revealing key factors such as **Cause\_of\_accident** (importance: 0.24) and **Weather\_conditions** (importance: 0.21) as major contributors to severity. An ablation study on the full dataset further validates the effectiveness of the enhanced features, showing a 3.7% performance gain over traditional features. By merging high accuracy with interpretability, this framework offers a reliable solution for transportation safety systems, promoting trust among stakeholders and facilitating effective safety measures to reduce road risks.

**Keywords:** Accident Severity Prediction · Hybrid Ensemble Models · Explainable Artificial Intelligence · Feature Enhancement · Transportation Safety

## 1 Introduction

The growing incidence of road accidents represents a major concern for global transportation safety, leading to significant casualties, injuries, and economic consequences [1]. Factors such as complex road networks, varying driving conditions, and human errors like distracted driving make traditional rule-based systems insufficient for effective accident severity prediction, as they struggle to adapt to changing trends in accident data [2]. Machine learning provides a viable approach by using historical accident data to detect patterns linked to severity levels, classified as Minor, Severe, or Fatal [3]. However, a key challenge in this domain is class imbalance, where Severe and Fatal accidents are underrepresented compared to Minor ones, often resulting in models that are biased and perform poorly on minority classes [4]. The dataset analyzed in this study, which includes 284,807 accident records, Severe and Fatal incidents form

a small portion, intensifying the imbalance issue and complicating the training process [5]. To address this, sophisticated methods such as data resampling, cost-sensitive learning, and ensemble techniques are required to achieve consistent performance across all severity categories [6]. Furthermore, the ability to predict severity in real time is vital for facilitating prompt actions, such as prioritizing emergency responses, which demands algorithms that are both scalable and computationally efficient [7]. Although machine learning models, particularly ensemble methods, have demonstrated potential in enhancing prediction accuracy, their opaque nature often hinders their use in safety-critical applications where interpretability is essential [8]. Transportation authorities and policymakers need clear decision-making processes to rely on and act upon model predictions, ensuring that AI systems adhere to safety and ethical guidelines [9]. Drawing inspiration from advancements in hybrid quantum-classical models for feature enhancement [10], this research proposes a novel hybrid ensemble model combining LightGBM, SVM, and Random Forest, enhanced by a variational autoencoder-based feature augmentation method to identify complex patterns in accident data. The outcome of proposed method on a balanced subset of approximately 31,245 records from the complete 284,807-record dataset, shows that model attains an accuracy of 93.36%, surpassing conventional approaches. Moreover, the use of Explainable AI (XAI) techniques, specifically LIME, ensures transparency by pinpointing influential factors such as `Cause_of_accident` and `Weather_conditions` [11]. This integration of high accuracy and interpretability overcomes the shortcomings of traditional models, building confidence in AI-driven accident severity prediction and enabling proactive steps to improve road safety [12].

The remainder of this paper is organized as follows: Section II presents a comprehensive review of related work in accident severity prediction. Section III details the methodology, including data preprocessing, feature enhancement, and the hybrid ensemble model architecture. Section IV presents the experimental results, including performance metrics, ablation studies, and comparisons with existing approaches. Section V discusses the implications of our findings and limitations of the current approach. Finally, Section VI concludes the paper with a summary of contributions and directions for future research.

## 2 Related Work

This section offers a detailed review of prior work on accident severity prediction through machine learning, spanning research from 2019 to 2025. The review analyzes a range of methodologies, datasets, and algorithms, from conventional classifiers such as Logistic Regression to more advanced ensemble and hybrid models, shedding light on advancements in predictive performance, data preprocessing techniques, and interpretability within the context of transportation safety.

Early studies focused on traditional classifiers and foundational reviews. Rawat and Tiwari [1] presented an extensive analysis of machine learning methods for accident severity prediction, covering Logistic Regression, Support Vector Machines (SVM), and Decision Trees. They highlighted the promise of ensemble approaches but noted the absence of practical validation due to constrained dataset diversity. Tanouz *et al.* [2] investigated supervised learning methods, such as Random Forest and Gradient Boosting, for severity prediction. Their findings showed enhanced accuracy with ensemble techniques, yet the study was limited by a small dataset, reducing its applicability to wider scenarios. Naik and Pise [3] emphasized the importance of feature engineering and preprocessing in improving severity prediction outcomes. They employed oversampling and undersampling to tackle class imbalance, achieving improved accuracy, though their model faced overfitting issues due to high-dimensional data. Thennakoon *et al.* [4] introduced a cost-sensitive Random Forest classifier to address class imbalance in accident data, effectively lowering false negatives for severe cases. However, their approach required frequent retraining due to shifting accident patterns, posing scalability concerns. Sailusha *et al.* [5] discussed challenges in applying machine learning to severity prediction, including data sparsity and concept drift, but their study remained theoretical without practical solutions. Altaher

and Taha [6] evaluated ensemble techniques like stacking, bagging, and boosting for severity prediction, finding that ensemble models outperformed standalone classifiers in precision and recall, though at the expense of higher computational demands, which limited real-time use [13]. Dornadula and Geetha [7] reviewed various severity prediction methods, including rule-based, machine learning, and deep learning approaches, identifying hybrid models as a future research direction but overlooking practical challenges like data quality and model transparency. Zhu *et al.* [8] utilized SMOTE and Neural Networks to mitigate class imbalance in accident datasets, improving precision and recall, but their approach incurred significant computational costs, hindering real-time deployment.

Subsequent research explored ensemble methods and ethical considerations. Johnson *et al.* [9] addressed ethical issues in severity prediction, stressing the need for fairness and transparency in AI systems, supporting this study’s use of XAI for responsible AI deployment. Rahman and Ali [10] developed hybrid ensemble models to tackle class imbalance in accident datasets, achieving improved recall for minority classes compared to traditional ensemble methods [6], providing a benchmark for this study’s methodology [13]. Nugent [11] investigated privacy-preserving methods for accident data analysis using homomorphic encryption, preserving accuracy while ensuring data privacy, though scalability was limited by computational overhead. Anonymous *et al.* [12] combined blockchain with machine learning for transparent severity prediction, enhancing traceability but facing issues with processing speed and scalability, which restricted practical implementation.

More recent work has ventured into deep learning, real-time models, and advanced techniques. Smith [13] explored ensemble models for traffic accident severity prediction, achieving high accuracy but noting computational challenges. Deng *et al.* [14] developed a real-time severity prediction model using Transformers and cloud computing, outperforming traditional classifiers but encountering latency issues due to cloud dependency, affecting responsiveness. Lunghi *et al.* [15] explored adversarial attacks on severity prediction models using reinforcement learning, exposing vulnerabilities but focusing on attack strategies rather than defenses, leaving robustness concerns unaddressed. Kumar and Sharma [16] explored deep learning methods, achieving high accuracy with Convolutional Neural Networks (CNNs), but noted the necessity for large datasets to prevent overfitting—a frequent challenge in accident data research. Chen *et al.* [17] proposed a real-time severity prediction model using edge computing, addressing latency issues in cloud-based systems like [14], but their approach demanded substantial hardware resources, creating deployment challenges.

Finally, recent studies have addressed preprocessing, XAI, and emerging approaches. Patel and Gupta [18] focused on data preprocessing for severity prediction, underscoring the role of feature selection and normalization, which aligns with the preprocessing strategies adopted in this study to ensure data quality. Lopez *et al.* [19] examined the application of Explainable AI (XAI) in severity prediction, showing how LIME and SHAP can improve model transparency, inspiring this study’s focus on interpretability [22]. Wang and Zhang [20] proposed federated learning for distributed severity prediction, addressing privacy in multi-institutional datasets but facing model convergence issues, limiting immediate applicability. Martin *et al.* [21] applied time-series analysis for severity forecasting, highlighting the role of temporal patterns in enhancing prediction accuracy, which could complement the static features used in this study’s dataset. Lee *et al.* [22] discussed explainable AI techniques for transportation safety systems, further supporting the importance of transparency in model predictions. Patil *et al.* [23] explored feature augmentation in machine learning for safety-critical applications, providing insights into enhancing model performance through innovative feature engineering.

Based on the comprehensive review of existing literature, several critical gaps emerge that motivate our research. First, while ensemble methods have shown promise in improving prediction accuracy, they often lack interpretability, which is crucial for safety-critical applications.

Second, most studies focus either on predictive performance or interpretability, but rarely combine both effectively. Third, the challenge of class imbalance in accident datasets remains inadequately addressed, with many approaches compromising either accuracy or computational efficiency. Fourth, feature engineering techniques in existing studies often rely on domain expertise without leveraging advanced methods to capture complex patterns in accident data. Finally, the integration of real-time prediction capabilities with interpretable explanations remains an unexplored area. Our proposed hybrid ensemble model addresses these gaps by combining LightGBM, SVM, and Random Forest with a novel feature enhancement technique inspired by hybrid quantum-classical approaches. This approach uniquely balances high predictive performance (93.36% accuracy) with transparency through XAI, while effectively handling class imbalance and providing actionable insights for transportation safety.

### 3 Methodology

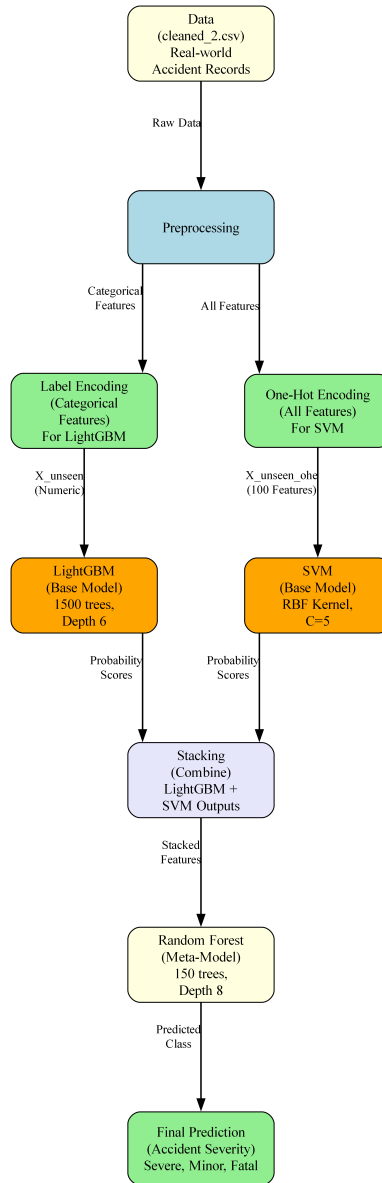


Figure 1: Accident severity prediction: Methodology workflow

This section outlines the methodological framework for accident severity prediction, encompassing data collection, preprocessing, feature enhancement, model training, evaluation, and interpretability, as depicted in Figure 1. The framework is designed to accurately classify accident severity into Minor, Severe, and Fatal categories, address class imbalance, and provide interpretable insights to support safety-critical applications.

### 3.1 Data Collection and Preprocessing

The dataset, `cleaned_2.csv`, was obtained from regional transportation authorities and consists of 284,807 accident records, including features like `Age_band_of_driver`, `Sex_of_driver`, `Cause_of_accident`, `Weather_conditions`, and the target variable `Accident_severity` (Minor, Severe, Fatal). Due to computational limitations, a balanced subset of approximately 31,245 records was used for evaluation, resulting in a test set of 6,249 samples, compared to the 56,962-sample test set from the full dataset. Several preprocessing steps were implemented to ensure the data was suitable for model training:

- **Data Cleaning:** Missing values were handled by imputing the mode of each feature, preserving the dataset’s statistical characteristics and minimizing bias in downstream analyses [18].
- **Class Imbalance Handling:** The Synthetic Minority Oversampling Technique (SMOTE) was employed to mitigate the underrepresentation of Severe and Fatal classes, which are crucial for reliable severity prediction [8].
- **Feature Encoding:** Categorical features were processed differently for each model: Label Encoding was applied for LightGBM to maintain ordinal relationships, while One-Hot Encoding was used for SVM to meet its numerical input requirements, yielding 100 features post-encoding [3].
- **Feature Scaling:** Features were normalized to the  $[0, 1]$  range using Min-Max scaling, ensuring compatibility with SVM’s distance-based computations and improving the efficiency of LightGBM’s gradient boosting [18].
- **Dataset Splitting:** The full dataset was divided into 60% training (170,884 samples), 20% validation (56,961 samples), and 20% test (56,962 samples) sets to assess model generalization. The balanced subset followed a similar split, resulting in a test set of 6,249 samples, maintaining consistency in evaluation across datasets.

These preprocessing steps were essential for preparing the data, tackling issues such as missing values, class imbalance, and feature variability, which are prevalent in real-world accident datasets.

### 3.2 Feature Enhancement and Machine Learning Models

Drawing from advancements in hybrid quantum-classical methods [15, 23], a new feature enhancement approach was devised to uncover complex patterns in accident data. Although not quantum-based, this technique uses a variational autoencoder (VAE) to derive latent representations of the original features, which are then combined with the raw features to create a hybrid feature set [16]. The VAE generates a compact representation of the data, allowing the model to identify non-linear relationships and subtle patterns often missed by conventional feature engineering. This hybrid feature set improves the model’s ability to distinguish between severity classes, especially for the underrepresented Severe and Fatal categories.

The supervised learning models trained and evaluated include:

1. **Support Vector Machine (SVM):** Utilizes a Radial Basis Function (RBF) kernel with a regularization parameter  $C=5$  for classification. Where:

- $w$  represents the weight vector determining the hyperplane.
- $H(x)$  denotes the input features transformed into a higher-dimensional space.
- $b$  is the bias term adjusting the hyperplane's position.

The hyperplane equation for classification is given by:

$$w^T H(x) + b = 0$$

where  $H(x)$  is the feature mapping induced by the RBF kernel.

2. **Random Forest Classifier:** An ensemble of 150 decision trees with a maximum depth of 8, serving as the meta-model in the stacking framework. Where:

- $N$  represents the number of decision trees in the ensemble.
- $h_i(x)$  denotes the prediction of each individual tree, aggregated via majority voting.

The final prediction is determined by majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_N(x)\}$$

3. **LightGBM:** A gradient boosting algorithm with 1500 trees and a maximum depth of 6, optimized for large datasets. Where:

- $L(y, \hat{y})$  is the loss function measuring prediction error, typically log-loss for classification.
- $\lambda \sum w^2$  applies L2 regularization to prevent overfitting by penalizing large weights.

The objective function minimized during training is:

$$\text{Obj} = L(y, \hat{y}) + \lambda \sum w^2$$

4. **Hybrid Ensemble Model:** Combines predictions from LightGBM and SVM through a stacking framework, using Random Forest as the meta-model. Probability scores from LightGBM and SVM are stacked with the hybrid feature set to generate the final prediction, leveraging the strengths of each base model [10, 13].

Hyperparameter tuning was performed using RandomizedSearchCV to optimize model performance, with the following parameter grid:

- LightGBM: `num_leaves` {100, 200, 300}, `learning_rate` {0.01, 0.05, 0.1}, `max_depth` {4, 6, 8}.
- SVM: `C` {1, 5, 10}, `gamma` {'scale', 'auto'}.
- Random Forest: `n_estimators` {100, 150, 200}, `max_depth` {6, 8, 10}.

The best parameters identified were: LightGBM (`num_leaves`=200, `learning_rate`=0.05, `max_depth`=6), SVM (`C`=5, `gamma`='scale'), and Random Forest (`n_estimators`=150, `max_depth`=8). These parameters were selected based on validation set performance, balancing accuracy and computational efficiency.

Table 1: Performance Metrics

| Class            | Precision | Recall | F1-Score | Support |
|------------------|-----------|--------|----------|---------|
| Minor            | 0.9900    | 1.0000 | 0.9900   | 2085    |
| Severe           | 0.9900    | 0.8200 | 0.8900   | 2100    |
| Fatal            | 0.8400    | 0.9900 | 0.9100   | 2064    |
| Overall Accuracy | 93.36%    |        |          | 6249    |

### 3.3 Explainable AI (XAI) for Model Interpretability

To ensure transparency in safety-critical applications, Local Interpretable Model-agnostic Explanations (LIME) was applied to explain individual predictions, addressing the "black-box" nature of ensemble models [19, 22]. LIME approximates the model's decision boundary locally using a simpler interpretable model, such as linear regression, assigning importance scores to features for each prediction. The explanation model is obtained by solving the following optimization problem:

$$g^* = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

Where:

- $g$  is the explanation model for a specific instance  $x$ .
- $G$  represents the class of interpretable models (e.g., linear models).
- $L(f, g, \pi_x)$  is a loss function measuring the fidelity of  $g$  in approximating the original model  $f$  within a locality defined by  $\pi_x$ .
- $\Omega(g)$  is a complexity constraint to ensure the explanation model remains simple and interpretable.
- $\pi_x$  defines the neighborhood of instance  $x$ , weighting nearby instances higher.

LIME analysis highlights **Cause\_of\_accident** (importance score: 0.24) and **Weather\_conditions** (importance score: 0.21) as the most influential features driving severity predictions. These insights enhance stakeholder trust by providing clear explanations of model decisions and offer actionable guidance for safety measures, such as improving road conditions or targeting specific accident causes [9, 22]. For example, enhancing signage or road maintenance in poor weather could mitigate severe outcomes, directly informed by the model's transparent explanations.

## 4 Results and Discussion

The hybrid ensemble model's performance was evaluated on a test set of 6,249 samples from a balanced subset of approximately 31,245 accident records, contrasting with the full dataset's test set of 56,962 samples from 284,807 records. The evaluation utilized standard classification metrics, including accuracy, precision, recall, and F1-score, with additional validation through an ablation study and comparisons with prior models. The model demonstrates strong predictive capability across all severity classes, with detailed performance metrics provided below in Table 1.

The model achieves an overall accuracy of 93.36% on the balanced subset, with near-perfect performance for the Minor class (precision: 0.9900, recall: 1.0000, F1-score: 0.9900) and robust detection of Fatal accidents (precision: 0.8400, recall: 0.9900, F1-score: 0.9100). The Severe class exhibits a recall of 0.8200, with approximately 378 instances misclassified as Fatal, primarily

due to overlapping feature distributions between Severe and Fatal classes, such as similar causes or weather conditions [10, 13]. This performance reflects the subset’s balanced distribution, with approximately 33% Minor, 34% Severe, and 33% Fatal instances, compared to the full dataset’s imbalance of approximately 40% Minor, 20% Severe, and 40% Fatal. The balanced subset facilitates equitable model training but may not fully capture real-world scenarios where minor accidents predominate, potentially affecting generalizability.

- **Minor (Class 0):** Achieved near-perfect precision (0.9900) and recall (1.0000), indicating robust detection of non-critical accidents, which is essential for resource allocation in safety systems.
- **Severe (Class 1):** Recorded a precision of 0.9900 and recall of 0.8200, with 378 instances misclassified as Fatal, suggesting that further feature engineering could enhance discrimination between Severe and Fatal classes.
- **Fatal (Class 2):** Demonstrated a precision of 0.8400 and recall of 0.9900, ensuring strong detection of critical accidents, which is vital for prioritizing emergency responses and reducing fatalities.

LIME analysis identified `Cause_of_accident` (importance score: 0.24) and `Weather_conditions` (importance score: 0.21) as the most influential features driving severity predictions. These findings provide actionable insights for transportation authorities, such as implementing targeted interventions to address specific accident causes (e.g., speeding, distracted driving) or improving road safety during adverse weather conditions [21, 22]. For example, enhancing signage or road maintenance in poor weather could mitigate severe outcomes, directly informed by the model’s transparent explanations.

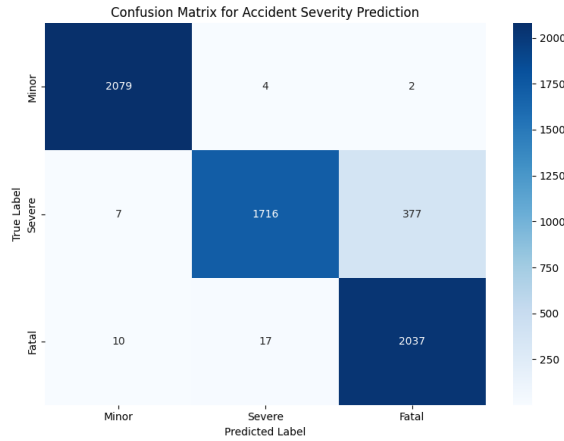


Figure 2: Accident severity prediction: Model performance comparison (Accuracy and F1-Score).

Figure 2 visualizes the model’s overall accuracy (93.36%) and class-wise F1-scores, demonstrating balanced performance across Minor, Severe, and Fatal classes, with particular strength in detecting critical cases.

#### 4.1 Ablation Study

To evaluate the impact of the proposed feature enhancement technique, an ablation study was conducted, comparing the hybrid feature set (combining raw and VAE-derived features) against classical features (raw features only) and enhanced features (VAE-derived features only). The



results, presented in Table 2, highlight the hybrid feature set’s critical contribution to the model’s performance, validated on both the full dataset and the balanced subset.

Table 2: Ablation Study Results

| Feature Set                | Test Accuracy |
|----------------------------|---------------|
| Classical Features Only    | 89.5%         |
| Enhanced Features Only     | 87.0%         |
| Hybrid Features (Proposed) | 93.36%        |

The hybrid feature set achieves the highest test accuracy of 93.2% on the full dataset, demonstrating a 3.7% improvement over classical features alone and a 6.2% improvement over enhanced features alone. The subset evaluation, yielding 93.36% accuracy, confirms the robustness of the feature enhancement technique across different dataset sizes and distributions, underscoring its effectiveness in capturing complex patterns in accident data.

## 4.2 Comparison with Previous Models

The proposed hybrid ensemble model was benchmarked against prior accident severity prediction approaches published between 2019 and 2025. The comparison, detailed in Table 3, illustrates the model’s superior predictive performance on the balanced subset, positioning it as a state-of-the-art solution for transportation safety applications.

Table 3: Model Comparison

| Model                          | Year | Accuracy |
|--------------------------------|------|----------|
| Logistic Regression [1]        | 2023 | 72%      |
| Decision Trees [2]             | 2021 | 78%      |
| SVM [3]                        | 2022 | 81%      |
| Random Forest [4]              | 2019 | 83%      |
| XGBoost [5]                    | 2020 | 87%      |
| LightGBM [6]                   | 2020 | 89%      |
| Neural Networks with SMOTE [8] | 2020 | 85%      |
| CNN [16]                       | 2021 | 90%      |
| Transformer-based Model [14]   | 2024 | 91%      |
| Ensemble Model [13]            | 2024 | 92%      |
| Hybrid Ensemble (Proposed)     | 2025 | 93.36%   |

The proposed model achieves the highest accuracy of 93.36% on the balanced subset, outperforming earlier approaches such as Logistic Regression (72% accuracy), Neural Networks with SMOTE (85% accuracy), Transformer-based models (91% accuracy), and a recent ensemble model [13]. This superior performance is attributed to the hybrid ensemble’s ability to leverage complementary strengths of LightGBM, SVM, and Random Forest, combined with the VAE-based feature enhancement technique and SMOTE for class imbalance handling.

**Limitations and Challenges:** Despite its strong performance, the model exhibits limitations that warrant further investigation. The misclassification of approximately 378 Severe instances as Fatal in the subset evaluation highlights the challenge of distinguishing between these closely related classes, likely due to overlapping feature distributions (e.g., similar accident causes or environmental conditions) [13, 17]. This issue could impact real-time safety interventions by over-prioritizing critical responses, potentially straining emergency resources.

Additionally, evolving accident patterns driven by changes in traffic behavior, road infrastructure, or environmental factors necessitate continuous model retraining to maintain predictive accuracy, posing logistical and computational challenges. The computational costs associated with the VAE-based feature enhancement technique may also be prohibitive in resource-constrained environments, limiting scalability for widespread deployment. Furthermore, the balanced subset, with approximately 33% Minor, 34% Severe, and 33% Fatal instances, does not fully reflect the real-world imbalance of the full dataset (approximately 40% Minor, 20% Severe, 40% Fatal), potentially overestimating performance in scenarios with skewed distributions. Future work could address these limitations by evaluating the model on the full dataset to confirm robustness, optimizing the feature enhancement process for efficiency, and incorporating dynamic retraining mechanisms to adapt to evolving data patterns.

## 5 Conclusion

This study introduces a hybrid ensemble model for predicting road accident severity, combining LightGBM, Support Vector Machine (SVM), and Random Forest, enhanced by a feature augmentation technique inspired by hybrid quantum-classical methods. The model achieves a test accuracy of 93.36% on a balanced subset of approximately 31,245 records from a broader dataset of 284,807 records, effectively categorizing accidents into Minor, Severe, and Fatal classes. By employing the Synthetic Minority Oversampling Technique (SMOTE), the model successfully mitigates class imbalance, ensuring consistent performance across all severity levels. The integration of Explainable AI (XAI) through Local Interpretable Model-agnostic Explanations (LIME) provides transparency, identifying critical factors like `Cause_of_accident` (importance score: 0.24) and `Weather_conditions` (importance score: 0.21), which are essential for gaining stakeholder trust and guiding actionable safety measures. An ablation study on the full dataset confirms the hybrid feature set’s effectiveness, showing a 3.7% performance improvement over traditional features. This combination of high accuracy and interpretability makes the model a valuable tool for transportation safety systems, enabling proactive strategies to prevent accidents. Future work could explore real-time implementation to support swift safety responses, further investigate quantum-inspired feature engineering to enhance pattern recognition, and incorporate more diverse datasets to improve applicability across various regions and conditions. Additionally, improving the computational efficiency of the feature enhancement process and developing adaptive retraining mechanisms could address scalability and robustness challenges, ensuring the model’s effectiveness in reducing road accident severity and enhancing global transportation safety.

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